

Theorem 6.32 (Gram-Schmidt Procedure). *Let $v_0, \dots, v_{m-1}$ a linear independent list in V , Let $f_0 = v_0$, for any $k = 1, \dots, m-1$:*

$$f_k = v_k - \frac{\langle v_k, f_0 \rangle}{\|f_0\|^2} f_0 - \dots - \frac{\langle v_k, f_{k-1} \rangle}{\|f_{k-1}\|^2} f_{k-1}$$

For any $k = 0, \dots, m-1$, define $e_k = \frac{f_k}{\|f_k\|}$, then $e_0, \dots, e_{m-1}$ is orthonormal in V , and for any $k = 0, \dots, m-1$ we have:

$$\text{span}(v_0, \dots, v_{m-1}) = \text{span}(e_0, \dots, e_{m-1})$$

Proof. The best way to prove an theorem about algorithm is to understand what the algorithm does. Let's put this formula aside and rephrase what the algorithm doing.

We first need a helper function, say *orthogonalize*: given v, w that $v \neq \lambda w$, produce f_w such that v is orthogonal to f_w and $f_w \in \text{span}(v, w)$, this is done by $w - \frac{\langle w, v \rangle}{\|v\|^2} v$.

Suppose v, u are orthogonal, w is orthogonal to v , we will show that $\text{orthogonalize}(u, w)$ is orthogonal to v . This is trivial by calculating $\langle w - \frac{\langle w, u \rangle}{\|u\|^2} u, v \rangle = \langle w, v \rangle - \langle \frac{\langle w, u \rangle}{\|u\|^2} u, v \rangle = 0 - 0 = 0$.

Then, let $f_0, \dots, f_{k-1}$ an orthogonal list and $v_k \notin \text{span}(f_0, \dots, f_{k-1})$, for any $i = 0, \dots, k-1$, define:

$$o_i = \begin{cases} \text{orthogonalize}(f_0, v_k) & \text{if } i = 0 \\ \text{orthogonalize}(f_i, o_{i-1}) & \text{otherwise} \end{cases}$$

We induction on i and show that o_i is orthogonal to $f_0, \dots, f_i$

- Base($i = 0$): $o_0 = \text{orthogonalize}(f_0, v_k)$ and it is orthogonal to f_0 by the property of *orthogonalize*.
- Ind($i = n + 1$): $o_{n+1} = \text{orthogonalize}(f_{n+1}, o_n)$, thus o_{n+1} is orthogonal to $f_{n+1} = f_i$ by the property of *orthogonalize*.

We have $o_n = o_{i-1}$ is orthogonal to $f_0, \dots, f_{i-1}$, then for each $f_0, \dots, f_{i-1}$, we have o_i is orthogonal to $f_0, \dots, f_{i-1}$ respectively,

cause $o_i = \text{orthogonalize}(f_i, o_{i-1})$ and o_{i-1} is orthogonal to $f_0, \dots, f_{i-1}$.

Then o_{k-1} is orthogonal to $f_0, \dots, f_{k-1}$, we can unfold o_{k-1} and see:

$$o_{k-1} = ((v_k - \frac{\langle v_k, f_0 \rangle}{\|f_0\|^2} f_0) - \dots) - \frac{\langle v_k, f_{k-1} \rangle}{\|f_{k-1}\|^2} f_{k-1}$$

.

Then we normalize the orthogonal list to get a orthonormal list.

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